



Original software publication

ESCOX: A tool for skill and occupation extraction using LLMs from unstructured text



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ABSTRACT

ESCOX, also known as ESCOSkillExtractor, is an open-source, non-proprietary tool for identifying and classifying skills, skillsets, and occupations from job postings and general text. It utilizes the European Skills, Competences, Qualifications and Occupations (ESCO) taxonomy to structure extraction, addressing the need for taxonomy-aligned skill identification in unstructured labor market data. Developed within the SKILLAB EU Horizon project, ESCOX combines LLMs and text embeddings to map content to standardized categories. It offers a user-friendly graphical interface for researchers, educators, and HR professionals, supporting skills gap analysis, training, recruitment, and policy planning, and contributing to the development of a skills-based economy.

Code metadata

Current code version	v0.1.17
Permanent link to code/repository	https://github.com/SoftwareImpacts/SIMPAC-2025-97
Permanent link to Reproducible Capsule	
Legal Code License	MIT License
Code versioning system used	git
Software code languages, tools, and services used	Python, JavaScript, HTML, CSS. Additional tool: ESCO embeddings
Compilation requirements, operating environments & dependencies	Python 3.10+, flask, transformers, sentence-transformers, pytorch (cuda 12), scikit-learn, scipy, summa, pandas, networkx, waitress
Link to developer documentation/manual	https://github.com/KonstantinosPetrakis/esco-skill-extractor/tree/master/docs
Support email for questions	dkavargy@csd.auth.gr , petrakki@csd.auth.gr
Release platform	https://pypi.org/project/esco-skill-extractor/
Easy way to install in Python	<code>pip install esco-skill-extractor</code>

1. Introduction

Despite the growing volume of online job data, accurately identifying labor market skills remains challenging [1]. Traditional keyword-based methods fail to address the complexity of job descriptions, especially in aligning soft [2] and hard skills with standardized taxonomies like ESCO [3] and ISCO-08 [4]. As digital transformation reshapes the job market, intelligent tools are needed to extract structured skill data

from unstructured sources, improving education-employment alignment. Online Job Advertisements (OJAs) [5] help track labor trends but suffer from inconsistencies and “ghost jobs” [6]. As a result, in many professional domains, academic curricula often struggle to keep pace with rapidly evolving industry demands, leading to skill mismatches between graduates and employers. This issue is evident across fields such as Software Engineering (SE), marketing, and healthcare, where technological advancements and shifting workforce needs highlight the urgent need for more targeted upskilling initiatives.

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¹ <https://skillab-project.eu/>

To address these challenges, ESCOX was developed as part of the EU Horizon research project SKILLAB¹ [7,8], which aims to enhance skill matching across Europe. It leverages real-time extraction, classification, and recommendation mechanisms, powered by ESCO taxonomy embeddings and LLMs, to improve semantic precision in skill detection. Integrated into the SKILLAB platform, it supports policymakers, educators, and businesses by providing real-time labor market insights, bridging skill gaps, and identifying emerging trends. This lightweight yet versatile solution enables stakeholders to make data-driven decisions for workforce development and education planning.

To install ESCOX, you can directly use pip from PyPI² by running the following command: `pip install esco-skill-extractor`. This method is recommended for users who wish to use ESCOX as a pre-built tool, without interacting with the source code. For developers or contributors who wish to modify the codebase or customize their personal version of ESCOX, it is recommended to clone the GitHub repository. The repository can be cloned by running the following command: `git clone https://github.com/KonstantinosPetrakis/esco-skill-extractor.git`. After cloning the repository, you can navigate to the project directory and install the necessary dependencies by executing `pip install -r requirements.txt`. ESCOX includes essential datasets located in the `esco_skill_extractor/data` directory within the repository. This directory contains the essential data required to generate the initial embeddings, enabling comparisons between unstructured text and ESCO skills as well as ISCO occupations. These files serve as the foundation for matching textual information with standardized ESCO skills and occupations, ensuring accurate and consistent mapping throughout the process. Further detailed instructions are available in the documentation located in the `docs/` directory of the repository.³ Additionally, ESCOX can be used directly through a Google Colab notebook,⁴ offering a ready-to-use environment without requiring local installation.

ESCOX is a versatile tool for analyzing labor market trends across diverse user groups. Researchers use it to track emerging technical and soft skills, supporting studies on skills evolution and curriculum development. Policymakers leverage its structured outputs to identify regional skill gaps and design targeted training programs, while HR professionals refine job descriptions and corporate training. Unlike many existing tools, ESCOX offers a user-friendly, no-code interface for extracting both skills and occupations within a single framework, combining accuracy with practical usability for education, analysis, and decision-making.

2. Main contributions of ESCOX

2.1. Contribution 1: Dual extraction of skills and occupations

ESCOX introduces a novel capability to extract both skills and occupations from unstructured textual data, treating them as interdependent elements of labor market intelligence. It maps detected entities to standardized ESCO and ISCO-08 codes, bridging the gap between competencies and job functions. A key innovation introduced by ESCOX lies in its ability to perform *dual information extraction*, identifying not only skills but also occupations directly from unstructured textual content. While most existing tools focus solely on extracting skills—often treating occupations as secondary metadata—, ESCOX treats both entities as equally important and interrelated components of labor market intelligence. This functionality enables the tool to detect explicit or implicit occupational references in texts and map them to standardized

Table 1

Top ESCO skills identified from job postings.

ESCO skills	Count	Proportion
java (computer programming)	2,611	0.2767
sql	1,802	0.1915
devops	1,204	0.1277
work independently	952	0.1011
computer programming	900	0.0958
ensure client orientation	712	0.0745
agile development	650	0.0692
python (computer programming)	558	0.0585
communication	400	0.0426
think analytically	352	0.0372
solve problems	320	0.0319
sql server	258	0.0266
scala	200	0.0213
office software	151	0.0160
Total ESCO Count of Skills	7,400	

ISCO-08 codes. By doing so, it bridges the gap between “what someone can do” (skills) and “what someone is qualified for” (occupations) [9], creating a richer, more actionable representation of human capital.

To demonstrate the practical value of this dual extraction capability, we conducted a real-world case study involving job postings related to SE in various EU regions. After setting up a Python web crawler, we collected and pre-processed approximately 6500 OJAs from EURES (European Employment Services).⁵ These postings were categorized into four ISCO-08 occupational groups by domain experts affiliated with the EURES network: Web and Multimedia Developers, Software Developers, Systems Analysts, and Applications Programmers, all of which represent core roles within the SE domain. The job descriptions were then processed using the ESCOX tool to identify the embedded skills and occupations. The most frequently detected skills and occupations in this dataset are presented in Table 1 and Table 2, respectively. This case study serves as a concrete validation of ESCOX’s dual extraction mechanism and illustrates its potential for large-scale labor market analytics.

Table 1 presents the most prevalent ESCO skills extracted from job postings, ranked by frequency and proportion. The distribution highlights strong demand for technical competencies (e.g., Java, SQL, DevOps) alongside essential workplace skills (‘work independently’, ‘ensure client orientation’). The proportions reflect realistic labor market trends, with programming languages collectively dominating (Java: 27.7%, SQL: 19.2%) while interpersonal skills appear less frequently (communication: 4.3%). Total counts ($N = 7400$) suggest robust dataset coverage, though the underrepresentation of non-technical skills may reflect either sampling bias or industry-specific hiring priorities.

Table 2 presents the most frequently identified ISCO-08 occupations extracted from the analyzed job postings. The results highlight the prominence of roles such as ICT Business Analyst, Project Manager, and Computer Scientist, which together account for a substantial portion of the dataset. These occupations reflect core functions within the SE and digital services domain. The proportions illustrate a diverse occupational distribution, underscoring the capability of the ESCOX tool to recognize both technical and managerial roles from unstructured text. This distribution also reinforces the tool’s potential to inform curriculum design and workforce planning based on real-world labor market demands.

2.2. Contribution 2: ESCOX is open source, modular, and reproducible

The system is open-source, modular, and domain-agnostic, enabling application across diverse text sources (e.g., policy reports, research

² <https://pypi.org/project/esco-skill-extractor/>

³ https://github.com/KonstantinosPetrakis/esco-skill-extractor/blob/master/esco_skill_extractor/_main_.py#L76-L88

⁴ https://colab.research.google.com/drive/1ksrth-uYNsLnT5Q75V9kUaF6yVUBf_Ue

⁵ https://eures.europa.eu/jobseekers_en

Table 2

Top ISCO-08 occupations identified from job postings.

ISCO-08 Occupations	Count	Proportion
ICT Business Analyst	2,150	0.3521
Project Manager	1,450	0.2374
Computer Scientist	950	0.1556
Business Consultant	800	0.1310
Technical Communicator	650	0.1064
ICT Security Specialist	580	0.0950
Data Protection Officer	520	0.0852
Industrial Production Manager	450	0.0737
Training & Development Professional	420	0.0688
Computer Science Lecturer	380	0.0622
Specialized Services Manager	350	0.0573
ICT Documentation Manager	320	0.0524
Technical Sales Professional	300	0.0491
ICT System Administrator	280	0.0458
Communication Scientist	250	0.0409
Total Count of Occupations	6,100	

articles). For lengthy inputs, ESCOX integrates a summarization mechanism to enhance efficiency and extraction accuracy. The final output is a structured CSV containing ESCO-aligned skills or occupations, facilitating seamless integration into labor market analysis pipelines.

2.3. Contribution 3: ESCOX adaptability to textual databases

ESCOX extends beyond job advertisements, operating across diverse textual sources such as policy documents, research articles, and institutional reports. Its flexibility enables skill and occupation extraction in fields like healthcare, chemistry, and marketing. To handle long or unstructured documents, ESCOX integrates a summarization mechanism using the *summa* library,⁶ applied only when input length exceeds a threshold. This approach improves extraction precision by minimizing irrelevant content, supporting broader labor market insights and interdisciplinary applications.

2.4. Contribution 4: ESCOX is fast, scalable and memory-efficient

A central contribution of ESCOX is its ability to perform fast, scalable skill and occupation extraction from unstructured text, making it suitable for both research and production environments. To empirically validate this capability, we conducted benchmarking experiments on the same dataset of over 6500 OJAs discussed earlier, executed across eight computing environments with diverse hardware configurations, including both CPU-only and GPU-enabled setups. On a standard Intel i7-14700F CPU-based workstation without GPU acceleration, ESCOX completed processing in 472.49 s, achieving a throughput of approximately 13.78 postings per second. In practical terms, this means that ESCOX can extract skills and occupations from nearly 14 job descriptions every second, even without GPU support.

As shown in Table 3, ESCOX demonstrates consistent performance across all tested systems. While high-performance machines with GPUs such as the NVIDIA A100 and L4 achieved significantly faster runtimes, the tool also performs reliably on standard CPU-based setups, reaffirming its efficiency and scalability in varied deployment scenarios. Importantly, memory consumption remained modest across all environments. Once again, on a standard Intel i7-14700F CPU-only machine, memory usage ranged from 394.68 MB to 536.43 MB, and even on GPU-enabled systems, peak usage did not exceed 2 GB. Furthermore, the difference between initial and peak memory consumption was consistently small—typically under 150 MB—indicating that ESCOX maintains a stable and lightweight memory footprint throughout execution. This predictable behavior makes the tool well-suited for both large-scale applications and resource-constrained environments.

⁶ https://github.com/skillab-project/Skillab-Tracker/blob/master/api/tasks/_helpers.py#L239-L272

3. ESCOX architecture

ESCOX is a Python tool that extracts ESCO-aligned skills and ISCO-08 occupations from unstructured text Fig. 1 using a pre-trained transformer model. Input text is encoded into embeddings and matched against ESCO/ISCO vocabularies via cosine similarity [10], with results filtered by a user-defined threshold. First, it processes all the skills from the ESCO classification system by converting them into numerical vectors using a lightweight language model. These vectors capture the meaning of each skill in a mathematical form. When new text is provided, like a job description or a research sentence, it is also converted into a vector using the same model. Then, the system calculates how close the new text is to each ESCO skill by using cosine similarity — a measure that shows how similar two pieces of text are. If the similarity is high enough and passes a predefined threshold, the text is matched to the corresponding skill. This approach allows ESCOX to identify relevant skills from any unstructured text efficiently and accurately, even when the text and skill titles are not exactly the same.

The technical architecture of ESCOX is depicted in Fig. 2. It is organized into five interconnected components, namely the backend architecture, database, deployment, core functionalities and graphical user interface. Each component plays a distinct role in enabling skill and occupation extraction from unstructured text.

3.1. Back-end

The backend of ESCOX is built using a Flask-based (Python) API, which exposes two core endpoints (Fig. 3): one for skill extraction and another for occupation extraction. Each endpoint accepts an array of text strings via POST requests and returns an array of arrays, where each sub-array contains the detected skills or occupations corresponding to the input. A GET / endpoint serves the primary HTML interface.

The Flask API utilizes a Sentence Transformer model to compute semantic similarity between input text (e.g., job descriptions) and ESCO skill or occupation entries. By default, the system employs the *all-MiniLM-L6-v2* model,⁷ which provides a strong balance between performance and accuracy. In the latest version of ESCOX, the architecture has been extended to accept an optional model parameter, allowing users to specify alternative pre-trained Sentence Transformer models based on their own performance, domain-specific needs, or platform compatibility. A list of supported models and usage instructions is available in the public GitHub repository.⁸

To illustrate the trade-off between speed and model complexity, we conducted a runtime comparison between the default model (*all-MiniLM-L6-v2*) and one of the fastest but lower-performing alternatives from the Sentence Transformers library: the *average_word_embeddings_glove.6B.300d* model. As shown in Fig. 4, the GloVe-based model completed the extraction process significantly faster (89.07 s) than MiniLM (472.49 s) for the same dataset of 6500 job postings. While GloVe provides substantial speed improvements, its semantic accuracy is lower in benchmark evaluations,⁹ reaffirming the importance of model selection depending on task requirements.

Flask is deployed through Gunicorn, a WSGI HTTP server optimized for UNIX environments, monitored via a Unix socket within the project directory. To manage incoming web traffic and serve static files efficiently, Nginx acts as a reverse proxy in front of Gunicorn. Additionally, Cross-Origin Resource Sharing (CORS) is enabled in Flask to allow requests from different domains, which is essential when the frontend and backend are deployed separately.

⁷ <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

⁸ <https://github.com/KonstantinosPetrakis/esco-skill-extractor?tab=readme-ov-file#possible-keyword-arguments-for-skillextactor>

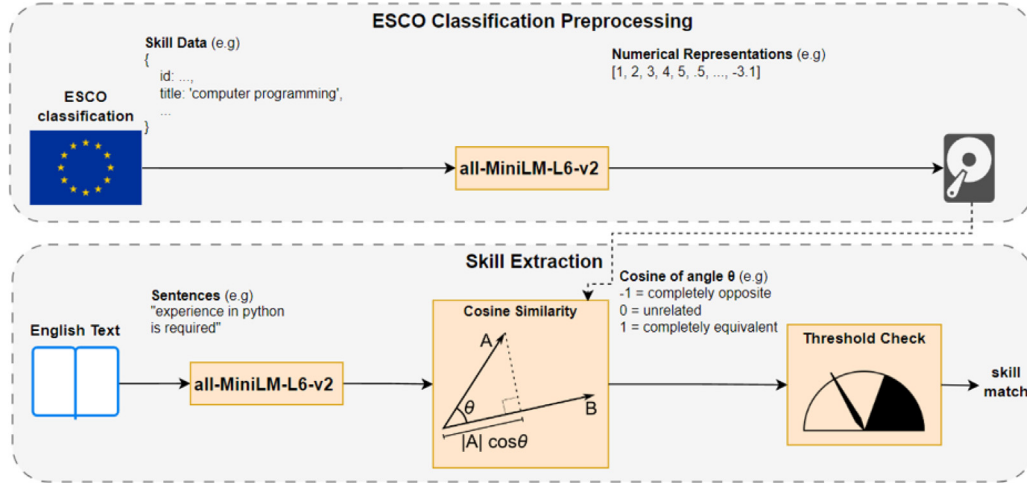
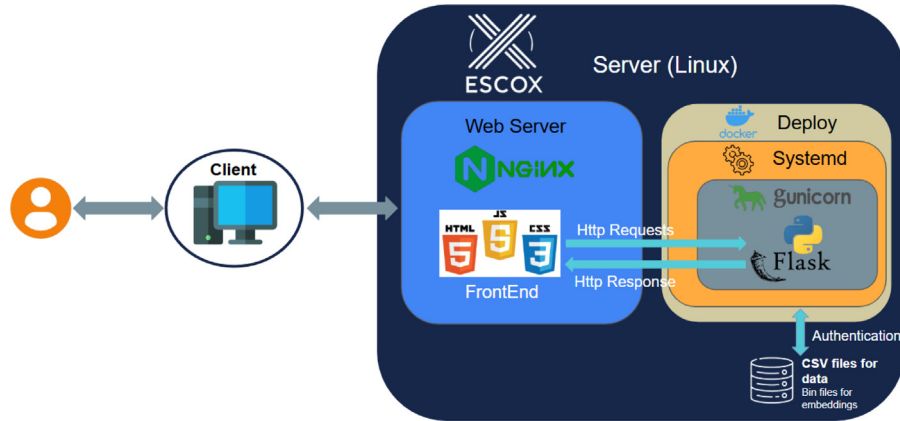
⁹ <https://huggingface.co/blog/mteb>

Table 3

Performance benchmarks of ESCOX across different machines.

Machine	CPU / RAM / GPU	Runtime (s)	Throughput (jobs/s)	Memory (MB)
1	Intel i7-14700F 2.10 GHz / 16GB / No GPU	472.49	13.78	394.7 → 536.4
2	Intel i7-13620H 4.90 GHz / 10GB / RTX 1060 8GB	948.39	6.87	455.92 → 462.34
3	Intel i7-8750H 2.20 GHz / 16GB / RTX 4060 4GB	608.94	10.70	885.44 → 944.02
4	Intel Xeon(R) 2.00 GHz / 51GB / 15GB	131.18	49.65	1847.25 → 1922.21
5	Intel Xeon(R) 2.20 GHz / 53GB / NVIDIA L4 22.5GB	130.73	49.82	1903.61 → 1979.60
6	Intel Xeon(R) 2.20 GHz / 83GB / NVIDIA A100-SXM4-40GB	136.62	47.67	1957.00 → 2043.60
7	AMD EPYC 9B14 / 170.31GB / No GPU	391.31	16.64	1802.54 → 1960.40
8	AMD EPYC 7B13 / 47.1GB / No GPU	526.41	12.37	1781.74 → 1893.54

Note: Memory (MB) usage is reported as initial → peak consumption during execution.

**Fig. 1.** System description of ESCOX.**Fig. 2.** ESCOX architecture. The schema is inspired by the Smyrida tool from Merkoureas et al. [11].

3.2. Database

ESCOX uses .CSV files to store the ESCO skills and occupations metadata and BIN files to store precomputed sentence embeddings. This design choice allows ESCOX to avoid recomputing embeddings at runtime, significantly speeding up response times during skill and occupation extraction.

3.3. Deployment

In production, ESCOX is typically deployed using Docker Compose to orchestrate services such as Flask, Gunicorn, and Nginx. Gunicorn manages the application processes, while Nginx acts as a reverse proxy, handling incoming HTTP requests, serving static files, and improving performance. While Docker Compose is recommended for production

deployments to manage services like Flask, Gunicorn, and Nginx efficiently, it remains optional during development. Developers may choose to run the application locally using a Python virtual environment and the built-in Flask development server, which provides faster feedback during code changes and simplifies the development workflow without containerization overhead.

3.4. Functionalities

ESCOX provides ten core functionalities (Table 4) to extract, process, and analyze skills and occupations from unstructured text. Each extracted skill or occupation is accompanied by a cosine similarity score ranging from 0 to 1, which serves as a proxy for confidence in the match. This score reflects the semantic closeness between the input text and ESCO entries, and can guide users in threshold calibration.

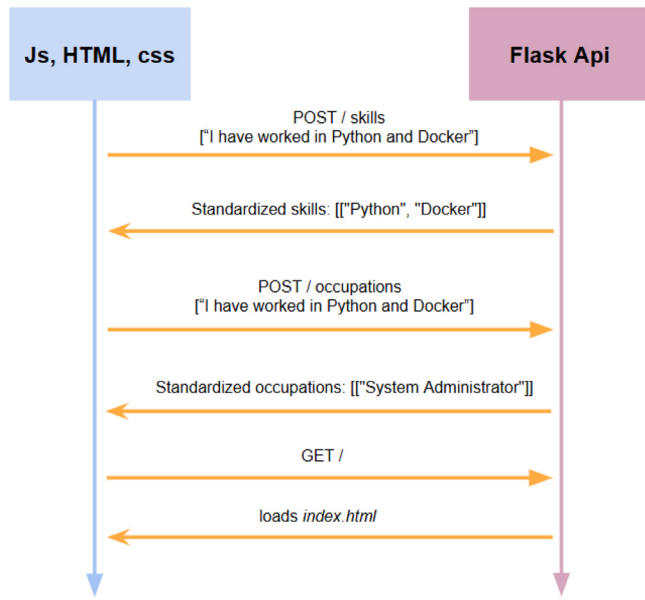


Fig. 3. Requests and responses between Frontend and Flask API.

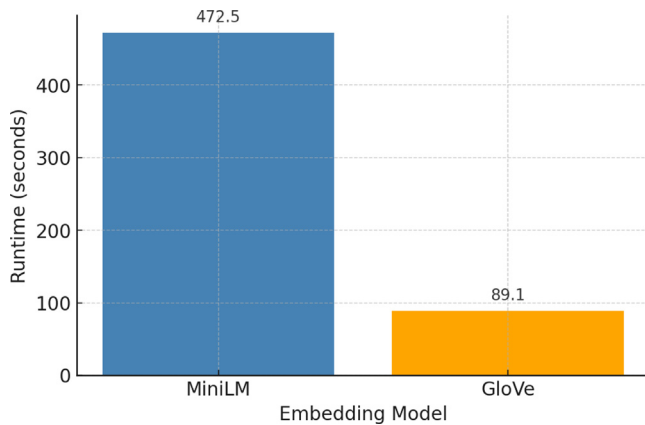


Fig. 4. Runtime comparison between all-MiniLM-L6-v2 and GloVe models.

These functionalities include text cleaning, summarization for long documents, threshold tuning for similarity matching, batch processing, and data export capabilities. The full set of functionalities is accessible through a simple API interface and designed to support both research and industrial applications.

3.5. Graphical user interface

ESCOX provides an intuitive, no-code interface that enables non-technical users — such as policymakers, HR professionals, educators, and analysts — to upload text, extract skills and occupations, and explore results through interactive visualizations. Its streamlined three-step workflow eliminates the need for programming, promoting wider adoption and simplifying the analysis process.

As illustrated in Fig. 5, ESCOX logo appears first, followed by the tool's workflow, which is organized into three simple steps. In the first step ①, users specify whether they wish to extract skills or occupations from the input text, depending on their analysis needs. In the second step ②, users provide job postings, research abstracts, or any other relevant textual data in raw text format, either by pasting it into the interface or uploading a file. In the third step ③, ESCOX automatically processes the input, applying semantic similarity to extract the

most relevant skills or occupations. Users can then easily review the extracted results, copy the detected skills, and directly match them with the official ESCO skill/occupation names through the interface for further analysis or integration into their projects.

4. Qualitative comparison

We present a qualitative analysis of state-of-the-art skill extraction tools compared to our ESCOX framework (Table 5). Evaluated tools include SkillNER,¹⁰ CareerBERT,¹¹ NESTA,¹² ESCOXML-R,¹³ SkillSpan,¹⁴ JobBERT,¹⁵ and SkillGPT.¹⁶ We assess each tool's taxonomy integration, extraction mechanism, domain adaptability, classification scope, interface/API availability, and licensing model.

Tool Analysis:

- **SkillNER**: Integrates EMSI taxonomy with Streamlit GUI; effective for structured CVs but limited domain adaptability.
- **CareerBERT**: Specializes in ESCO occupation classification without skill extraction or API support.
- **NESTA**: Customizable ESCO/Lightcast pipeline with web interface; focuses on broad skill profiling.
- **ESCOXML-R**: Multilingual XLM-RoBERTa model for ESCO tasks; API-only with open licensing.
- **SkillSpan**: BERT-based span extraction with annotated datasets; lacks taxonomy integration.
- **JobBERT**: Normalizes job titles using skill-augmented embeddings.
- **SkillGPT**: Combines ESCO integration with LLM-powered multilingual extraction.
- **ESCOX Framework**: Combines embedding-based ESCO matching with GUI/API access, domain adaptability through curated job postings, and MIT License openness. This positions ESCOX as a transparent, human-centered solution addressing technical and practical demands across academic, developer, and policy contexts while aligning with Horizon Europe priorities.

5. Illustrative example

We present one illustrative examples to showcase key functionalities of ESCOX tool. The example involves analyzing job postings from EURES to extract relevant skills and occupations.

5.1. Reviewing a single job posting

In this illustrative example (Table 6), we showcase how the skill extraction tool processes a real job posting retrieved from EURES. The position refers to a Software Engineer (Mid-Senior) role based in Patra, Greece. By parsing the job title and description, the tool identifies relevant ESCO skills such as Apache Maven, Python (computer programming), and Agile development. These skills are then mapped to standardized ISCO-08 occupations, including Digital Transformation Manager and System Analyst. To ensure the accuracy and differentiation of skills, we adjust the identified ESCO skills by mapping them to their corresponding ESCO URIs. This step minimizes the overlap between similar skills and ensures that each skill is uniquely identified. After this adjustment, the tool maps the URIs to the actual skill names, aligning with ESCO. This process demonstrates the ability

¹⁰ <https://github.com/nicolamelluso/SkillNER>

¹¹ <https://github.com/nickward/CareerBERT>

¹² https://github.com/nestauk/ojd_daps_skills

¹³ <https://github.com/mainlp/escoxlmr>

¹⁴ <https://github.com/kris927b/SkillSpan>

¹⁵ <https://github.com/jensjorisdecorde/JobBERT-evaluation-dataset>

¹⁶ <https://github.com/aida-ugent/SkillGPT>



Choose the entity to extract

Skills ☒ Occupations ☐

1

We are seeking a talented and motivated Data Analyst to join our GCS Aftermarket Services, Analytics and Fleet Operations department at Kongsberg Maritime. The successful candidate will play a crucial role in developing data-driven solutions to enhance energy efficiency and performance optimization for maritime operations.

In the Eco Insight team, we provide valuable insights from historical operational data to help our customers improve vessel energy efficiency and meet their sustainability targets. Our team offers analytics services for external deliveries, providing insights and advice on more energy-efficient operations. We also deliver insights to internal stakeholders on operational energy efficiency, supporting pre-upgrade, post-upgrade verification analysis, as well as application and product development.

Active communication and interaction with our customers are integral parts of the Eco Insight team's activities. We believe that maintaining strong relationships with our clients is essential for understanding their needs and delivering tailored solutions. Our team regularly engages with customers to gather feedback, discuss product progress, and ensure that our services align with

2

Extract skills

Copy CSV

- <http://data.europa.eu/esco/skill/05fc6f8d-639c-4329-a6de-dfdd800ed091>
- <http://data.europa.eu/esco/skill/082967d4-b94c-4305-aa13-35a6aa8c5c7c>
- <http://data.europa.eu/esco/skill/08e2ff26-88ff-47b5-9d37-d19ace01b075>
- <http://data.europa.eu/esco/skill/1605025e-a179-421f-8f35-5b07d182a6b2>
- <http://data.europa.eu/esco/skill/199f7919-5114-41b6-b6a5-41e0e4896ec1>
- <http://data.europa.eu/esco/skill/1fcce318-3212-487a-ac07-4941bc85060d>
- <http://data.europa.eu/esco/skill/20e7c937-a5e5-4a27-bb87-ddda5826f777>

3

Fig. 5. Graphic User Interface (GUI).

of the tool to interpret unstructured job data and provide structured outputs, which can be further used for labor market analysis, upskilling recommendations, or curriculum design.

6. Software impact

The widespread adoption of enterprise systems that generate large volumes of event logs opens up opportunities for leveraging software

tools that can automatically extract and analyze knowledge from these logs to enhance business process management. From a user's perspective, ESCOX offers a straightforward and accessible platform for extracting skills and occupations from job postings or resumes. The tool supports comparative analyses by providing different extraction models that can be adapted to various domains. Its intuitive interface allows for easy customization, enabling users to tailor the extraction process based on the specific skills and occupations they are interested

Table 4
ESCOX core functionalities.

Functionality	Description and implementation	Output
Skill Extraction	Extracts ESCO skills from text using the all-MiniLM-L6-v2 model. Compares input embeddings against precomputed ESCO skill embeddings.	List of matched ESCO skill IDs with descriptions (CSV/GUI)
Occupation Extraction	Identifies ISCO-08 occupations using the same embedding approach as skill extraction, with occupation-specific comparisons.	List of matched occupation IDs (CSV/GUI)
Threshold Configuration	Adjustable cosine similarity thresholds (default: 0.6 for skills, 0.55 for occupations) control matching strictness.	Filtered results based on user-defined thresholds
API Integration	RESTful API (Flask) with POST endpoints for automated text processing without GUI interaction.	Structured CSV response
Web Interface	Interactive GUI (JavaScript/HTML/CSS) with text input box and file upload capabilities.	Visual results display with clickable ESCO URIs
CSV Export	Generates downloadable CSV files containing extracted skills/occupations with full metadata.	Standardized CSV format for external analysis
Embedding Cache	Precomputed binary embeddings (BIN files) for all ESCO entities enable faster processing.	Reduced computation time after initial setup
Multilingual Support	Automatically detects and translates EU languages to English using integrated LLM before processing.	English output with optional source language preservation
Colab Support	Preconfigured Google Colab notebook with dependency management for zero-setup execution.	In-notebook visualization and analysis
SKILLAB Alignment	Maintains API and data structure compatibility with the SKILLAB Tracker platform.	Harmonized output ready for SKILLAB ingestion

Table 5
Qualitative comparison of skill extraction tools.

Tool	Taxonomy integration	Extraction mechanism	Domain adaptability	Classification scope	GUI availability	API support	Open-Source license
SkillNER [12]	Yes	Yes	No	Yes	Yes	Yes	Yes
CareerBERT [13]	Yes	Yes	No	No	Yes	Yes	Yes
NESTA [14]	Yes	Yes	Yes	No	Yes	Yes	Yes
SkillGPT [15]	No	Yes	Yes	Yes	Yes	Yes	Yes
ESCOXLM-R [16]	Yes	Yes	Yes	Yes	No	Yes	Yes
SkillSpan [17]	No	Yes	Yes	Yes	No	Yes	Yes
JobBERT [18]	Yes	Yes	Yes	Yes	No	Yes	Yes
Our approach (ESCOX)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

in. The integration of interactive visualizations in ESCOX plays a crucial role in enhancing the interpretability and explainability of the skill extraction process. By visualizing the extracted data, ESCOX helps users in rapidly identifying key insights, such as the most in-demand skills or occupational trends, enabling more informed decision-making. This renders ESCOX as a valuable tool for linking data analytics with practical skill management and workforce planning.

ESCOX is a web-based application with a user-friendly interface, making it accessible from anywhere as well as easy to use for those who may not have expertise in process mining. Unlike other open-source tools, which often require users to be familiar with the complexities of skill extraction, ESCOX is designed with simplicity in mind, ensuring that even non-experts can effectively utilize its capabilities. From a technical standpoint, ESCOX features a modular architecture, allowing for the integration of additional skill extraction functionalities, such as those needed for process discovery or conformance checking. This flexibility also sets the stage for developing and incorporating predictive process monitoring approaches, which typically leverage machine

learning algorithms. By incorporating human-in-the-loop mechanisms, ESCOX not only enhances data analytics but also ensures a more interactive and insightful experience. Furthermore, since ESCOX is primarily Python-based, it seamlessly integrates with various Python libraries, extending its capabilities and enabling easy adaptation and adjustments with new developments in the field.

Although this study presents the first published description of ESCOX, the tool has already been adopted in ongoing research projects for large-scale skill extraction and occupational mapping. These applications are currently under development and have not yet been formally published. In future versions of this manuscript and related publications, we intend to provide citations to both our own studies and other independent research efforts that employ ESCOX, once these works become publicly available.

7. Conclusions and future work

In this study, we introduce a methodology for the automatic extraction of both soft and technical skills from textual data. The proposed

Table 6
Job posting and related skills and occupations.

Field	Details
ID	1137
Organization	XXXX
Title	Software Engineer
Description	As a Software Engineer at XXXX, you will be at the forefront of innovation by designing, developing, and troubleshooting and debugging software programs. Your role is pivotal in developing software tools...(continued)
Experience Level	Mid-Senior
Type	Full-time
Location	Patra, Greece
Upload Date	2024-09-04
Source	EURES
Source ID	150 402
Mapped Skills	http://data.europa.eu/esco/skill/0d168770-4d9c-4096-b048-a6577818d306 - Apache Maven http://data.europa.eu/esco/skill/ccd0a1d9-afda-43d9-b901-96344886e14d - Python (computer programming) http://data.europa.eu/esco/skill/dba46f87-0831-49cd-a1c7-340a653c0221 - Agile development
Mapped Occupations	http://data.europa.eu/esco/occupation/719f101d-1866-49d3-8d5c-2256f606a8b9 - Digital Transformation Manager http://data.europa.eu/esco/isco/C2511 - Systems analysts

solution, ESCOX, is a supervised system trained on a curated scientific corpus annotated by domain experts. To facilitate broader adoption and usability, ESCOX has been developed as a web-based application, which can be deployed locally, making it accessible to users with limited technical expertise and without requiring prior programming knowledge, while maintaining a high level of accuracy. A key contribution of this work is the application of ESCOX to SE job profiles from the EURES database, utilizing the ESCO taxonomy for classification. Despite its promising results, there are some areas that require further exploration.

One potential limitation is that the accuracy of ESCOX could be improved by integrating more advanced deep learning architectures, such as transformer-based models like BERT, which have shown significant improvements in natural language processing (NLP) tasks, particularly Named Entity Recognition (NER). Although ESCOX currently relies on embedding-based methods for skill extraction, integrating fine-tuned transformers could further enhance its performance, especially in extracting soft and context-dependent skills. However, this would require training BERT or similar models on a domain-specific corpus, which is beyond the scope of this study. Another limitation is related to the training process of ESCOX, as it could be enhanced to handle larger datasets, given the increasing volume of job postings. As industries evolve and new skills emerge, expanding the training dataset with more diverse and contextually rich job descriptions could improve the model's adaptability.

Looking ahead, we plan to further enhance ESCOX's performance, transparency, and applicability in real-world settings. One direction involves integrating Approximate Nearest Neighbor (ANN) search algorithms [19]—such as those provided by FAISS—to significantly improve retrieval speed in large-scale deployments. In this context, ANN refers to a class of algorithms that enable efficient similarity search over high-dimensional vectors, while FAISS serves as a high-performance library implementing these techniques at scale. FAISS, developed by Meta, supports fast similarity search and clustering of dense vectors, making it ideal for handling high-volume labor market data.¹⁷ We also aim to extend the system's interpretability by incorporating advanced uncertainty estimation techniques. These include dynamic thresholding based on similarity score distributions and interactive visualizations to flag borderline or ambiguous matches. Inspired by recent advances in uncertainty-aware modeling [20,21], such enhancements can help users better assess the confidence of

predictions and adjust the system to their specific domain needs. Together, these future improvements are designed not only to optimize speed and reliability, but also to empower more informed decision-making in downstream applications such as training recommendations, occupational transitions, or personalized upskilling pathways.

CRediT authorship contribution statement

Dimitrios Christos Kavargyris: Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Konstantinos Georgiou:** Writing – review & editing, Validation, Data curation. **Eleanna Papaioannou:** Visualization, Data curation. **Konstantinos Petrakis:** Validation, Software, Resources. **Nikolaos Mittas:** Validation, Supervision. **Lefteris Angelis:** Project administration, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dimitrios Christos Kavargyris reports financial support was provided by Horizon Europe. Konstantinos Georgiou reports financial support was provided by Horizon Europe. Eleanna Papaioannou reports was provided by Horizon Europe. Konstantinos Petrakis reports financial support was provided by Horizon Europe. Nikolaos Mittas reports financial support was provided by Horizon Europe. Lefteris Angelis reports financial support was provided by Horizon Europe. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

A supplementary video is provided to demonstrate the functionality of ESCOX tool, showcasing the user interface and the skill extraction process using the ESCO taxonomy. The video can be accessed at [this link](#).

¹⁷ <https://github.com/facebookresearch/faiss>

References

- [1] W. Wan, C.-B. Ge, M.C. Friesen, S.J. Locke, D.E. Russ, I. Burstyn, S. Peters, Automated coding of job descriptions from a general population study: Overview of existing tools, their application and comparison, *Ann. Work. Expo. Heal.* 67 (5) (2023) 663–672.
- [2] K. Kechagias, Teaching and Assessing Soft Skills, SCST, 2011.
- [3] M. le Vrang, A. Papantoniou, E. Pauwels, P. Fannes, D. Vandestein, J. De Smedt, ESCO: Boosting job matching in Europe with semantic interoperability, *Computer* 47 (10) (2014) 57–64.
- [4] S. Kurt, W. Sermeus, ISCO-08 Classification and scope of practice of nurses: Survey among WHO-Europe member states, *Eur. J. Pub. Health* 34 (Supplement_3) (2024) ckae144–1828.
- [5] A.P. Carnevale, T. Jayasundera, D. Repnikov, Understanding Online Job Ads Data: A Technical Report, Tech. Rep., Georgetown University, Center on Education and the Workforce, 2014.
- [6] H. Ng, Why is it so hard to find a job now? Enter Ghost Jobs, 2024, arXiv preprint arXiv:2410.21771.
- [7] M. Aluas, L. Angelis, I. Arapakis, E. Arvanitou, K. Georgiou, A. Gogos, V. Voulgarakis, SKILLAB: Skills matter, in: 2024 50th Euromicro Conference on Software Engineering and Advanced Applications, SEAA, IEEE, 2024, pp. 491–498.
- [8] K. Georgiou, R. Rossini, M. Jahn, D. Tsoukalas, V. Voulgarakis, S. Tsekeridou, L. Angelis, SKILLAB: Creating a skills supply and demand data space, in: Proceedings of the 4th Eclipse Security, AI, Architecture and Modelling Conference on Data Space, 2024, pp. 10–17.
- [9] A. Levenson, C. Zoghi, Occupations, human capital and skills, *J. Labor Res.* 31 (4) (2010) 365–386.
- [10] A.R. Lahitani, A.E. Permanasari, N.A. Setiawan, Cosine similarity to determine similarity measure: Study case in online essay assessment, in: 2016 4th International Conference on Cyber and IT Service Management, IEEE, 2016, pp. 1–6.
- [11] I. Merkouras, A. Kaoui, G. Theodoropoulou, A. Bousdekis, A. Voulodimos, G. Miaoulis, Smyrida: A web application for process mining and interactive visualization, *SoftwareX* 22 (2023) 101327.
- [12] S. Fareri, N. Melluso, F. Chiarello, G. Fantoni, SkillNER: Mining and mapping soft skills from any text, *Expert Syst. Appl.* 184 (2021) 115544.
- [13] J. Rosenberger, L. Wolfrum, S. Weinzierl, M. Kraus, P. Zschech, CareerBERT: Matching resumes to ESCO jobs in a shared embedding space for generic job recommendations, *Expert Syst. Appl.* (2025) 127043.
- [14] Nesta, A UK skills taxonomy, 2018, (Accessed 12 May 2023).
- [15] N. Li, B. Kang, T.D. Bie, SkillGPT: a RESTful API service for skill extraction and standardization using a Large Language Model, 2023, arXiv preprint arXiv:2304.11060.
- [16] M. Zhang, R. van der Goot, B. Plank, ESCOXML-R: Multilingual taxonomy-driven pre-training for the job market domain, in: Proceedings of the Annual Meeting of the Association for Computational Linguistics, ACL, 2023.
- [17] M. Zhang, K.N. Jensen, S.D. Sonniks, B. Plank, SkillSpan: Hard and soft skill extraction from english job postings, 2022, arXiv preprint arXiv:2204.12811.
- [18] J.-J. Decorte, J. Van Haute, T. Demeester, C. Develder, JobBERT: Understanding job titles through skills, in: Proceedings of the Workshop on Forecasting and Understanding Skill and Talent Demand (FEAST) At ECML-PKDD 2021, 2021.
- [19] Y.A. Malkov, D.A. Yashunin, Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs, *IEEE Trans. Pattern Anal. Mach. Intell.* 42 (4) (2018) 824–836, <http://dx.doi.org/10.1109/TPAMI.2018.2889473>.
- [20] J.M. Dolezal, A. Srisuwananukorn, D. Karpeyev, S. Ramesh, S. Kochanny, B. Cody, A.T. Pearson, Uncertainty-informed deep learning models enable high-confidence predictions for digital histopathology, *Nat. Commun.* 13 (1) (2022) 6572, <http://dx.doi.org/10.1038/s41467-022-34161-2>.
- [21] B. Lakshminarayanan, A. Pritzel, C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in: Advances in Neural Information Processing Systems, vol. 30, Curran Associates, Inc., 2017.